accompanying code is that it would help you hit the ground running with your own projects, as the Bayesian optimization framework is very general and "plug-and-play." The exercises are also helpful in this regard.

Generally, I hope this book will be useful to your machine learning needs and overall a fun read. Before we dive into the actual content, let's take some time to motivate the problem that Bayesian optimization sets out to solve.

1.1 Finding the optimum of an expensive, black-box function is a difficult problem

As mentioned above, hyperparameter tuning in ML is one of the most common applications of Bayesian optimization. We will explore this problem, as well as a couple of others, in this section as an example of the general problem of black-box optimization. This will help us understand why Bayesian optimization is needed.

1.1.1 Hyperparameter tuning as an example of an expensive black-box optimization problem

Say you want to train a neural network on a large data set, but you are not sure how many layers this neural net should have. You know that the architecture of a neural net is a make-or-break factor in deep learning, so you perform some initial testing and obtain the results shown in Table 1.1.

Table 1.1 An example of a hyperparameter tuning task. Our task is to decide how many layers the neural network should have in the next trial in the search for the highest accuracy. It's difficult to decide which number we should try next.

Number of layers	Accuracy on the test set
5	0.72
10	0.81
20	0.75

The best accuracy you have found, 81%, is good, but you think you can do better with a different number of layers. Unfortunately, your boss has set a deadline for you to finish implementing the model. And since training a neural net on your large data set takes several days, you only have a few trials remaining before you have to decide how many layers your network should have. With that in mind, you want to know what other values you should try so that you can find the number of layers giving the highest possible accuracy.

This task is typically called *hyperparameter tuning* in ML, where you want to find the best setting (hyperparameter values) for your model so as to optimize some performance metric such as predictive accuracy. In our example, the hyperparameter of our neural net is its depth (the number of layers). If you are working with a decision tree, common hyperparameters are the

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